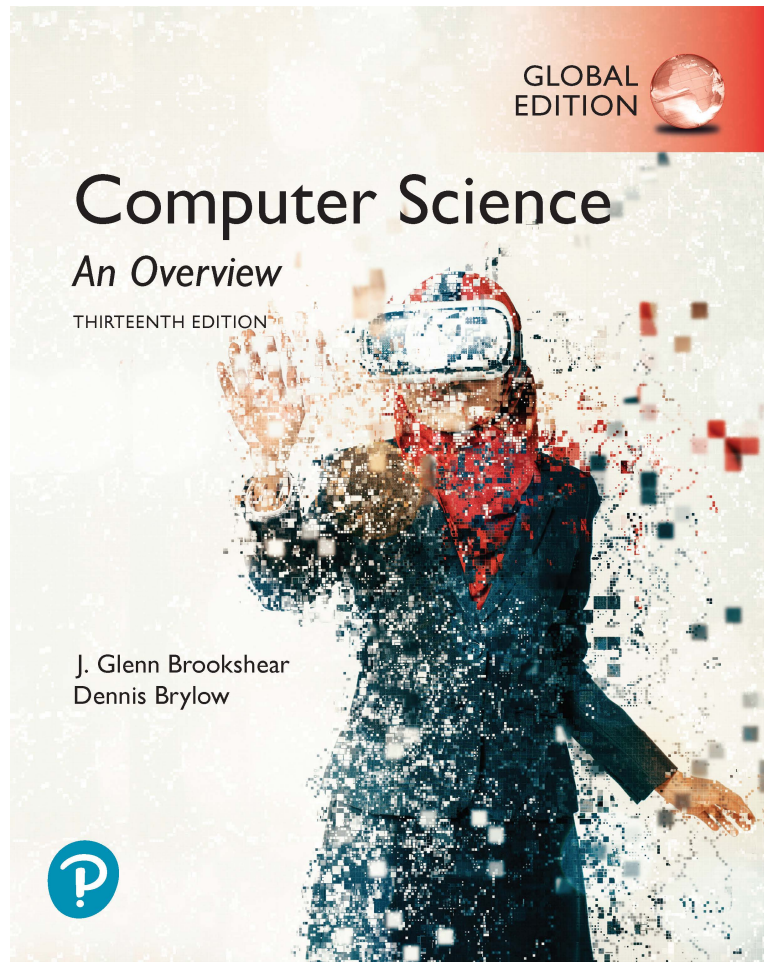


Computer Science An Overview

13th Edition, Global Edition



Chapter 4 Networking and the Internet

Chapter 4: Networking and the Internet

- 4.1 Network Fundamentals 基礎
- 4.2 The Internet
- 4.3 The World Wide Web
- 4.4 Internet Protocols 網路協定
- 4.5 Simple Client Server 主從式架構
- 4.6 Security

4.1 Network Fundamentals

- Network Software allows users to exchange information and share resources
 - Content
 - Software
 - Data storage facilities 數據存儲設施
- Network software has evolved into a network-wide operating system

Network Classifications

- Scope 範圍
 - Personal Area Network (short-range) 個人區域網路 (PAN)
 - Local Area Network (building/campus) 本地區域網路 (LAN)
 - Metropolitan Area Network (community) 都會區域網路 (MAN)
 - Wide Area Network (greater distances) 廣域網路 (WAN)
- Ownership 所有權
 - Closed versus open
封閉式 開放式
- Topology (configuration) 拓撲 (配置)
 - Bus (Ethernet) 乙太網路 P.5
 - Star (Wireless networks with central Access Point)
擁有一個中央存取點的無線網路!

Figure 4.1 Two popular network topologies (1 of 2)

有線結構的拓撲網路配置

a. Bus

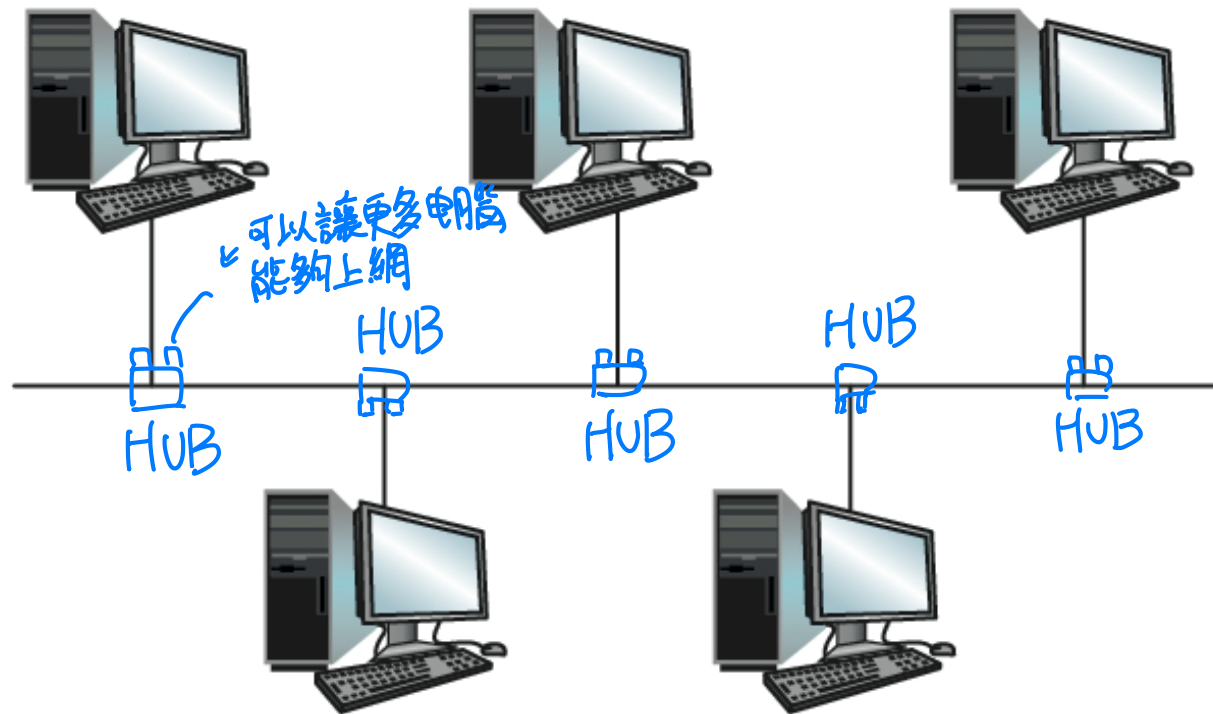
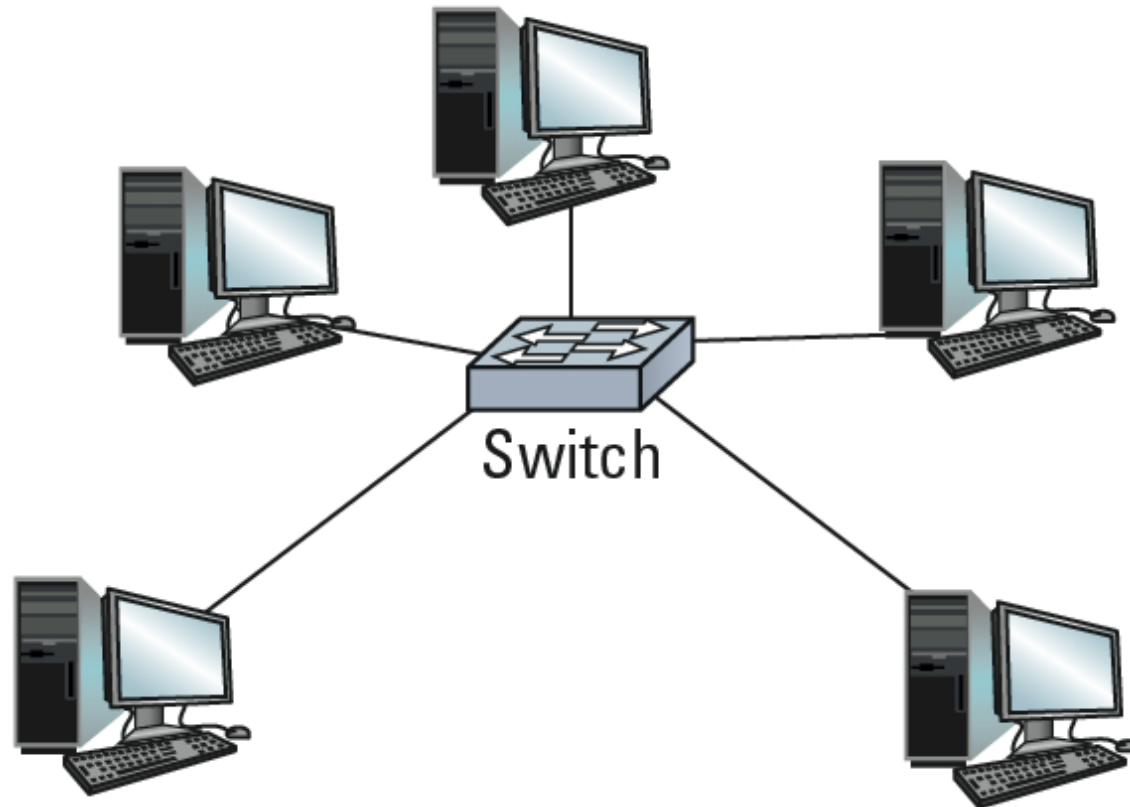


Figure 4.1 Two popular network topologies

(2 of 2)

無線結構的拓撲網路結構

b. Star



Protocols 網路協定

— "在網路上進行活動的規則"

- Rules by which activities are conducted on a network
- Example: Coordinating the transmission of messages between computers → 協調計算機之間的訊息傳輸
- Need to avoid all machines transmitting at the same time
* 需要避免所有機器同時傳輸
- Allows vendors to build products that are compatible with products from other vendors

常見協定：TCP、UDP、HTTP、HTTPS

TCP/IP 特点

「三向交握」

Three-way-Handshake

Protocols for Transmitting Messages

- CSMA/Collision Detection *CSMA / CD* *Carrier-Sence Multiple Access with Collision Detection*

- used in Ethernet

乙太網路所使用的協定

- both machines stop and wait for a independent, random time

*→ 若發生碰撞 或 該 Bus 不為 Silent
則雙方都等待一隨機的時間*

- CSMA/Collision Avoidance

無線網路所使用的協定

- used in WiFi, where not all machines can hear each other (hidden terminal problem)

由於用於 wifi, 因此不是所有機器都有辦法聽到訊号

- give advantage to the machine that has already been waiting

*產生
(Hidden Terminal)
Problem*

CSMA / CA *Carrier-Sence Multiple Access with Collision Avoidence*
載波偵聽多路訪問 / 碰撞避免

Figure 4.2 Communication over a bus network

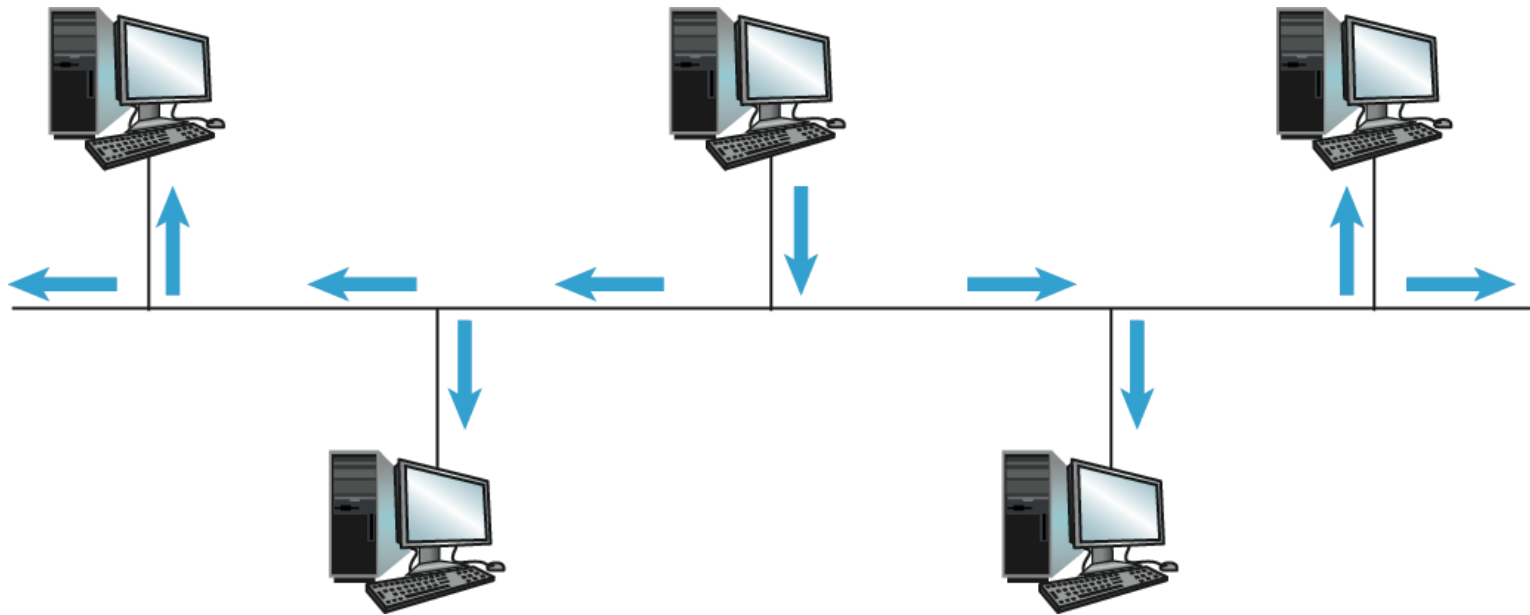
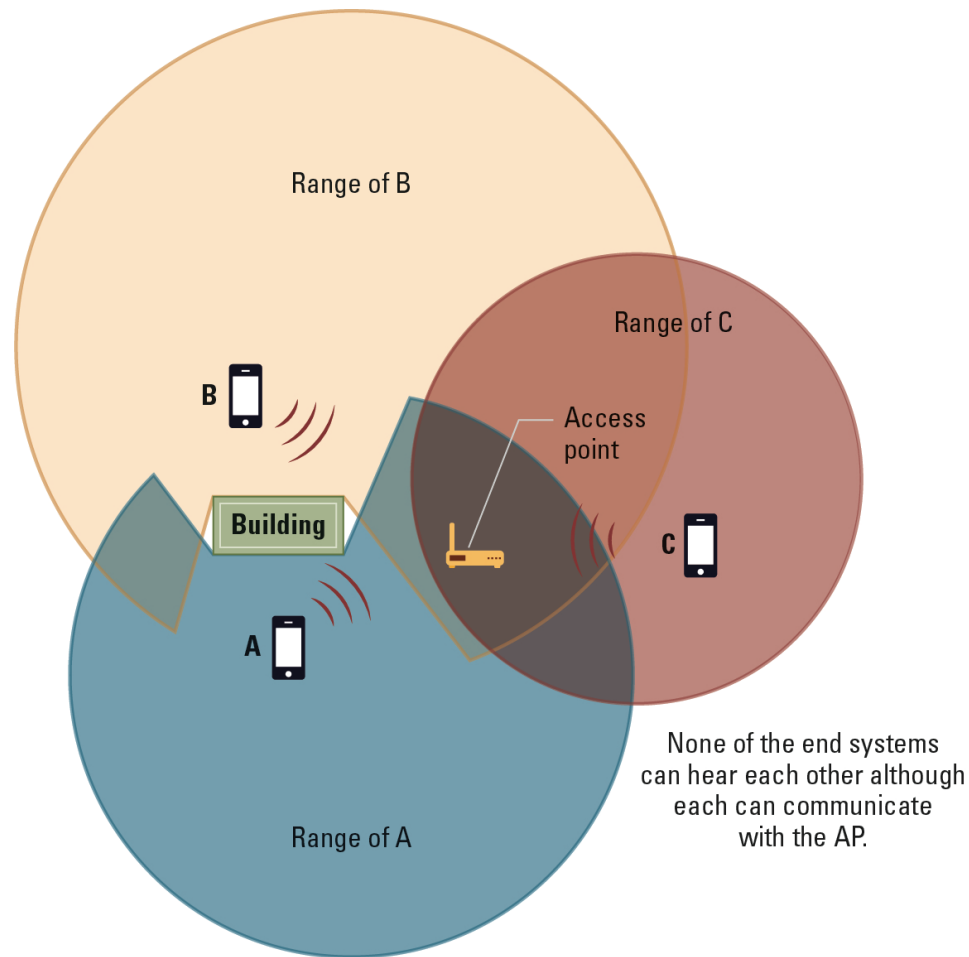


Figure 4.3 The hidden terminal problem



Combining Networks 構成網路的設備

延伸器、放大器

／ 將所有訊息都丟給另一邊的 Bus (無判斷)

- **Repeater:** passes all messages across two busses

橋接器

- **Bridge:** passes only messages that are destined for computers on the other bus

只將訊息目的地在另一個 Bus 的電腦的訊息轉送 (有判斷)

交換器

- **Switch:** acts like a bridge, but with connections to multiple busses

與 bridge 類似，但不同於 bridge 只能連接兩個 Bus，switch 能連接多個 bus。

路由器

- **Router:** Connects two incompatible networks

✧

重要

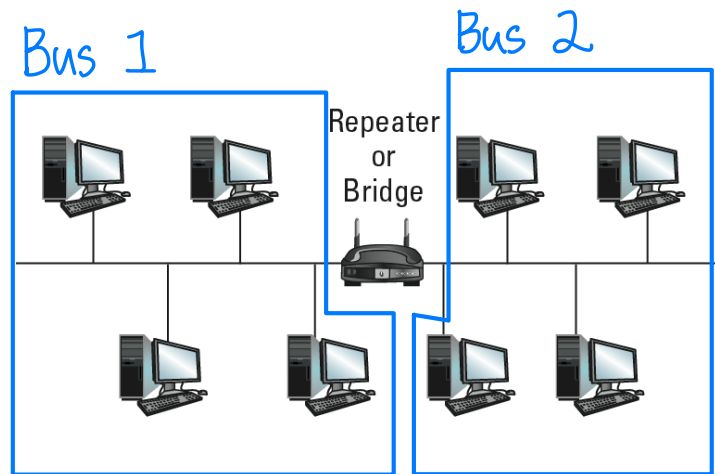
resulting in a network of networks called an **internet**

／ 連接兩個不相容的網路 (e.g. wifi & Ethernet)

使其能夠互連。

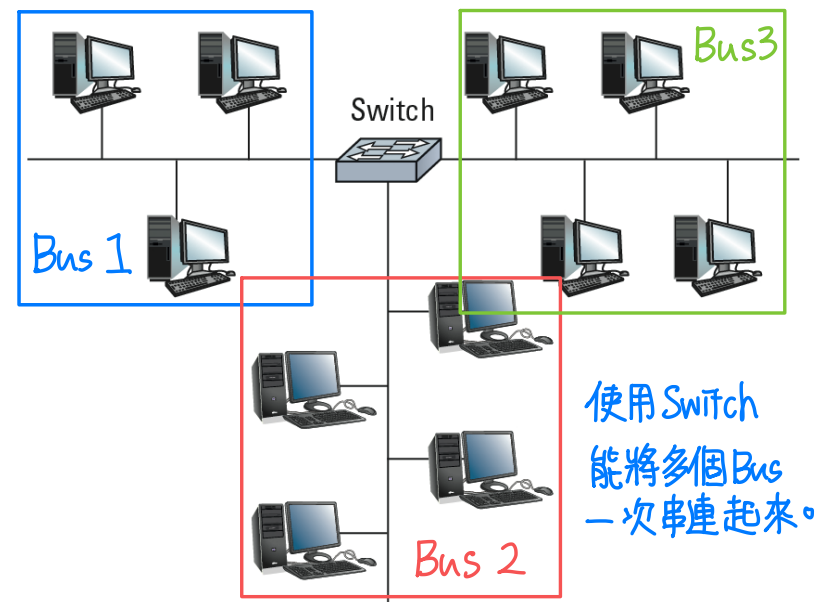
Gateway 門道 (網關)：一個網路連接到網際網路的點。因為其充當網路與外界的通道。

Figure 4.4 Building a large bus network from smaller ones



若要將 Bus 1、Bus 2 連接在一起
需要使用 Repeater 或者 Switch。

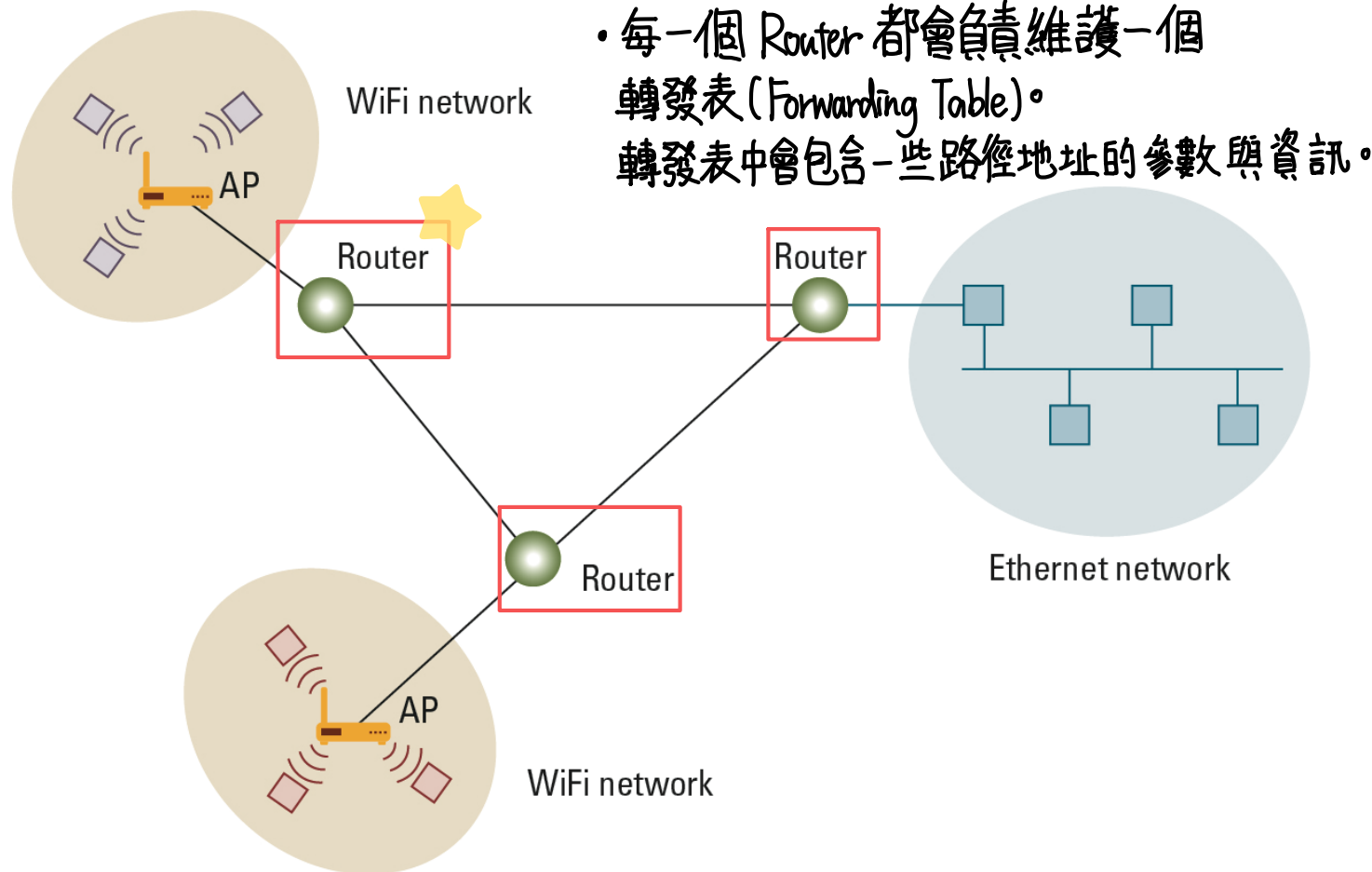
a. A repeater or bridge connecting two buses



使用 Switch
能將多個 Bus
一次串連起來。

b. A switch connecting multiple buses

Figure 4.5 Routers connecting two WiFi networks and an Ethernet network to form an internet



- 每一個 Router 都會負責維護一個轉發表 (Forwarding Table)。
轉發表中會包含一些路徑地址的參數與資訊。

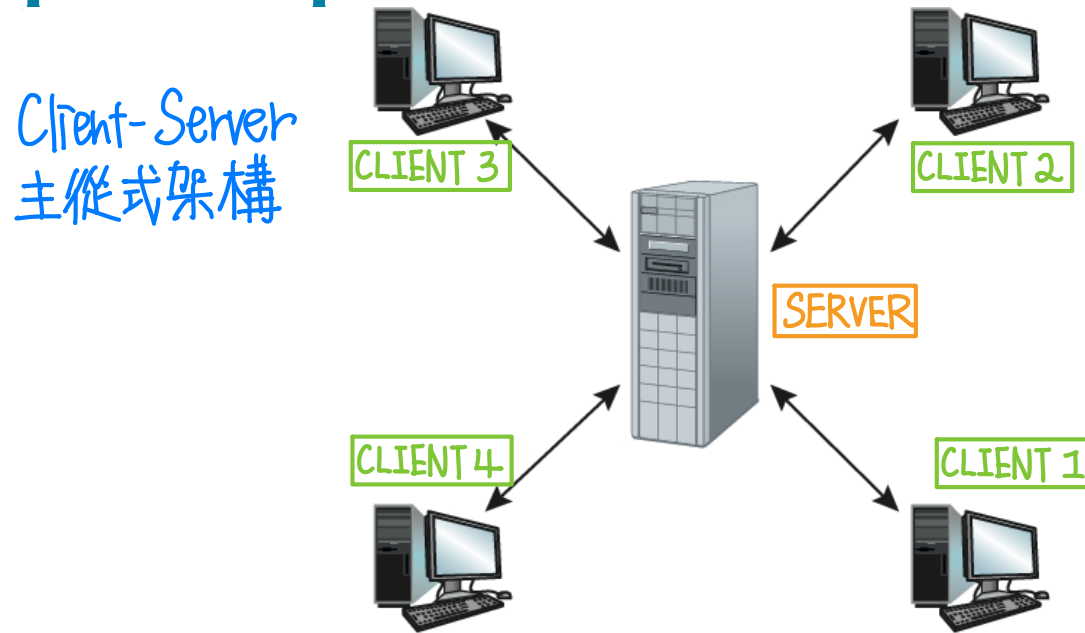
通訊方法

Methods of Process Communication

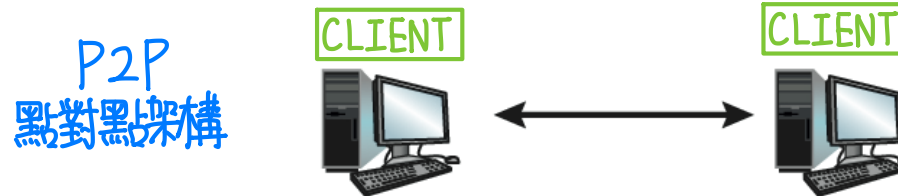
- Client-server 主從式架構 連續執行
 - Many clients, one server (executing continuously)
 - Clients make requests of other processes
 - Server satisfies requests made by clients
- Peer-to-peer (P2P) 点对点架構 (無主從式關係, 去中心化)
 - Two processes communicating as equals 兩個進程平等通信
 - Processes execute on a temporary basis

相對主從關係佔用較大的頻寬 (資源來自許多人)
硬碟寫入量較大。

Figure 4.6 The client/server model compared to the peer-to-peer model



a. Server must be prepared to serve multiple clients at any time.



b. Peers communicate as equals on a one-to-one basis.

相較集中式系統，
其沒有主要的管理者

Distributed Systems 分散式系統

- Systems units that execute as processes on different computers
 - Cluster computing 集群計算 (叢集計算)
 - Independent computers work closely together instead of a single, much larger machine 多台電腦 密切的相互合作
 - 以量取利 Grid computing 網格計算
 - Millions of home PCs (not connected to each other) work on a complex problem 沒有与 Cluster Computing 一樣 那麼緊密合作
 - Cloud computing 雲端計算
 - Provide services, hide the details 只負責提供服務、不提供任何其他底層資訊。

STOP ←
At Here!

大-上計概考試範圍!

4.2 The Internet 網路 UwU

- The Internet is an internet that spans the world
 - Original goal was to link a variety of networks into a connected system unaffected by local disasters
 - Today, it is a commercial undertaking that links a worldwide combination of PANs, LANs, MANs, and WANs involving millions of computers

<Review>

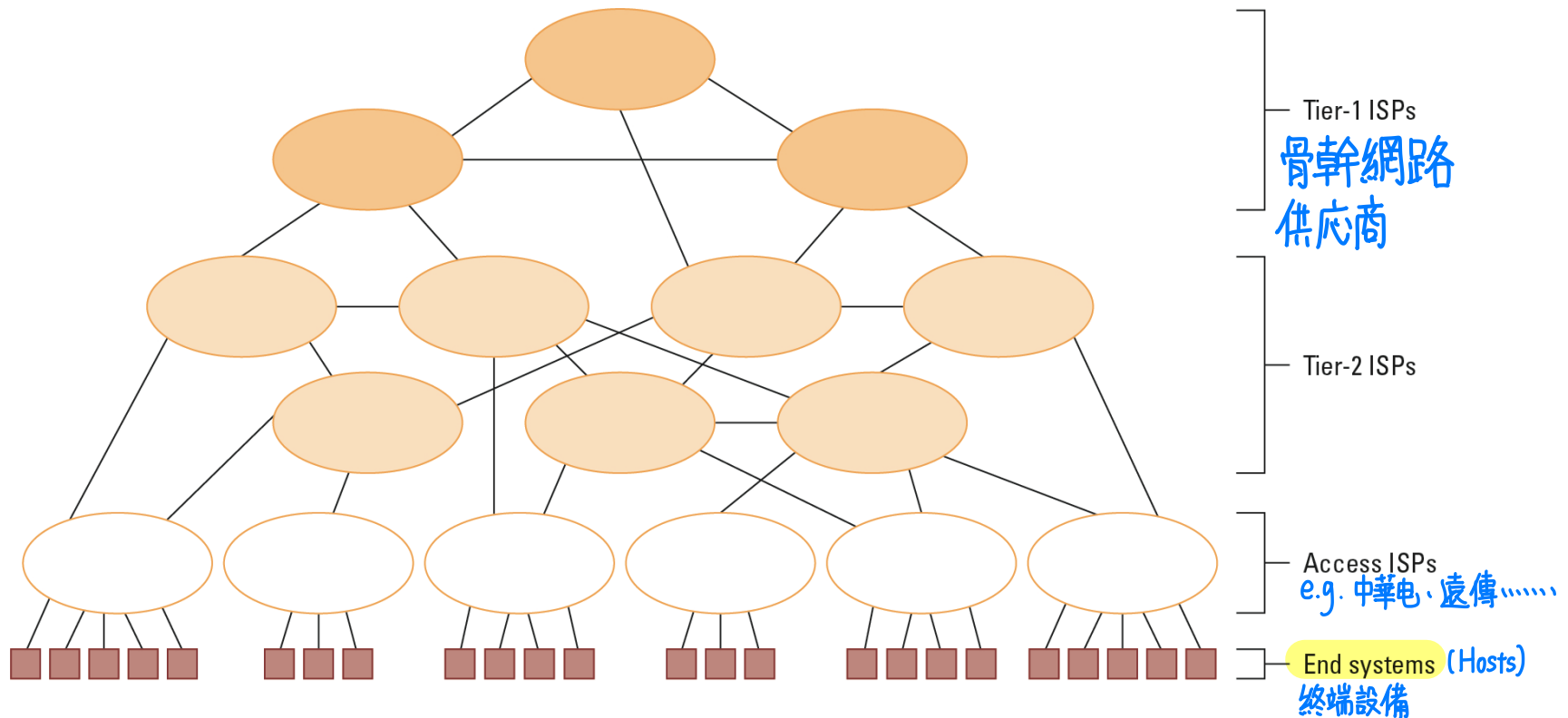
Router (路由器) 对網路的组成贡献 极高 AWA

期末從這裡开始!

Internet Architecture 網路架構

- Internet Service Provider (ISP) 網路服務供應商
 - Tier-1 (Internet backbone)
 - Tier-2 (骨幹網路)
 - Access or Tier-3 ISP: Provides connectivity to the Internet
 - Hot spot (wireless) 無線熱點
 - Telephone lines 電話線網路
 - Cellular 蜂窩網路、蜂巢網路
 - Cable/Satellite systems 有線網路、衛星網路
(與他人共享!)
e.g.
銅軸混合、
光纖……
- < Link 課本 P. 210 >
Intranet 企業網路
(內部網路)

Figure 4.7 Internet Composition



Internet Addressing 網路地址

IPv4 IPv6

- IP address: pattern of 32 or 128 bits often represented in dotted decimal notation
點分十進位法
- Mnemonic address: (助記地址)
為解決 IPv4 地址
不夠分配的問題
- 域名 Domain names (mu.edu)
- 頂級域名 Top-Level Domains
 - .org, .gov, .com, .mil, .net, .au, .ca, .biz,
- Domain name system (DNS)
 - Name servers 名稱伺服器
 - DNS lookup

Internet Corporation for Assigned Names & Numbers (ICANN)

網際網路名稱與數位位址分配機構

負責將IP位置區塊分配給各國的ISP,各國ISP再分配給使用者

- Allocates blocks of IP addresses to ISPs who then assign those addresses within their regions.
- Oversees the registration of domains and domain names.

Early Internet Applications 網路協定 (早期)

- Network News Transfer Protocol (NNTP) 網路新聞傳輸協定
- File Transfer Protocol (FTP)
- 明文 Telnet and Secure Shell (SSH)
- Hypertext Transfer Protocol (HTTP) 明文傳送
可被劫持
- Electronic Mail (email) 先將電郵傳送至 mail-server, 再由 mail server 傳送給 Receiver.
 - Domain mail server collects incoming mail and transmits outgoing mail
 - Mail server delivers collected incoming mail to clients via POP3 or IMAP 整合 POP3 & Web-mail 整合, Local 端可以对 web-mail 操作
Local 端
可以從 Mail Server Download 郵件, 載完後 server 端的檔案便會消失

- Sender

- Receiver

* 此處的颜色和課本剛好相反。

SMTP Simple Mail Transfer Protocol

220 mail.tardis.edu SMTP Sendmail Gallifrey-1.0; Fri, 23 Aug 2413 14:34:10

HELO mail.skaro.gov

250 mail.tardis.edu Hello mail.skaro.gov, pleased to meet you

MAIL From: dalek@skaro.gov

250 2.1.0 dalek@skaro.gov... Sender ok → checking whether in blacklist or not

RCPT To: doctor@tardis.edu

250 2.1.5 doctor@tardis.edu... Recipient ok → checking the receiver's mail box
can receive mail or not.

DATA

354 Enter mail, end with "." on a line by itself

Subject: Extermination.

EXTERMINATE!

Regards, Dalek

.

250 2.0.0 r7NJYAEI028071 Message accepted for delivery

QUIT

221 2.0.0 mail.tardis.edu closing connection

More Recent Applications

基於IP的语音傳輸

• Voice Over IP (VoIP) 利用網際網路協定來通訊，
達成语音通话、多媒体會議。

• Internet Multimedia Streaming 網路多媒体串流 (应用範例: 下載時同時觀看)

— 單播 N-unicast Unicast: 一个 Sender 傳訊息給一个 Receiver

— 多播 Multicast N-unicast: 一个 Sender 被包在多个 unicast 中
(e.g. 一个人傳訊息給多个人)

— On-demand streaming (实例应用: 中華電信MOD)

— Content delivery networks (CDNs)

內容分發網路 (实例应用: Cloudflare CDN)
賽博活佛

補 Anycast 任播:

允許任意 user 連線至最近
的 Server

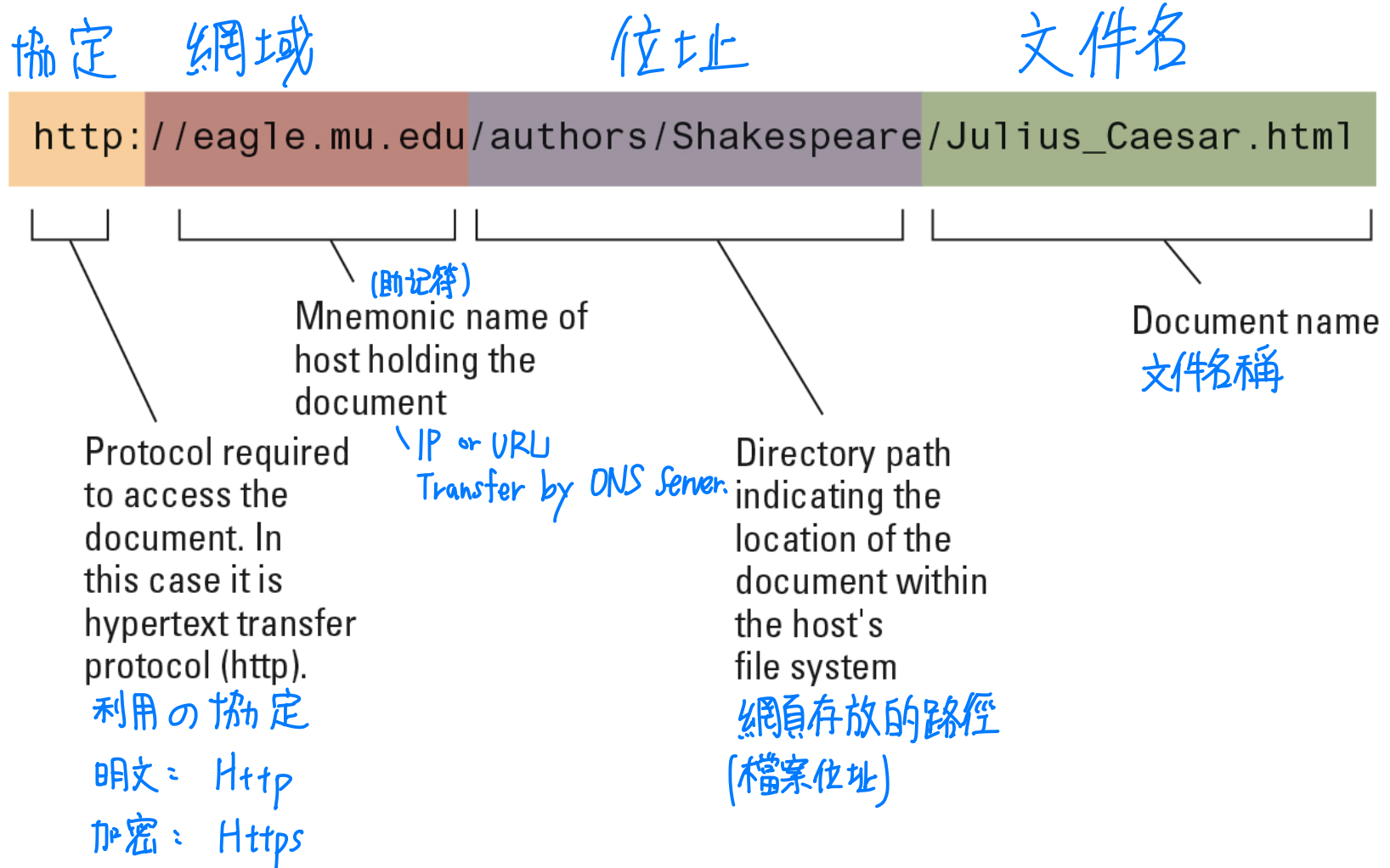
多播 Multicast: 一个 Server 藉由 Router 傳遞訊息給多个 Client
一个 address, 然後藉由 Router 來轉送。

4.3 World Wide Web

- **Hypertext** combines internet technology with concept of linked-documents
 - Embeds **hyperlinks** to other documents
超連結
- **Browsers** present materials to the user 瀏覽器呈現 HyperText
- **Webservers** provide access to documents Webserver 提供內容 (HyperText)
- Documents are identified by **URLs** and transferred using **HTTP**
Uniform Resource Locator (URL)
網頁網址

協定+網域+位址+文件名 → 網址！

Figure 4.8 A typical URL



→ 超文本標記語言 (Not A Programming Language!!!)

Hypertext Markup Language (HTML)

- Encoded as text file
 - Contains tags to communicate with browser
 - Appearance 外觀相關的Tag
 - `<h1>` to start a level one heading
h1~h6
 - `<p>` to start a new paragraph
段落
 - Links to other documents and content (連結至其他連結或文件)
 - `<a href = . . . >`
 - Insert images 插入圖片
 - `<img src = . . . >`
- 还有很多

Figure 4.9 A simple webpage

a. The page encoded using HTML.

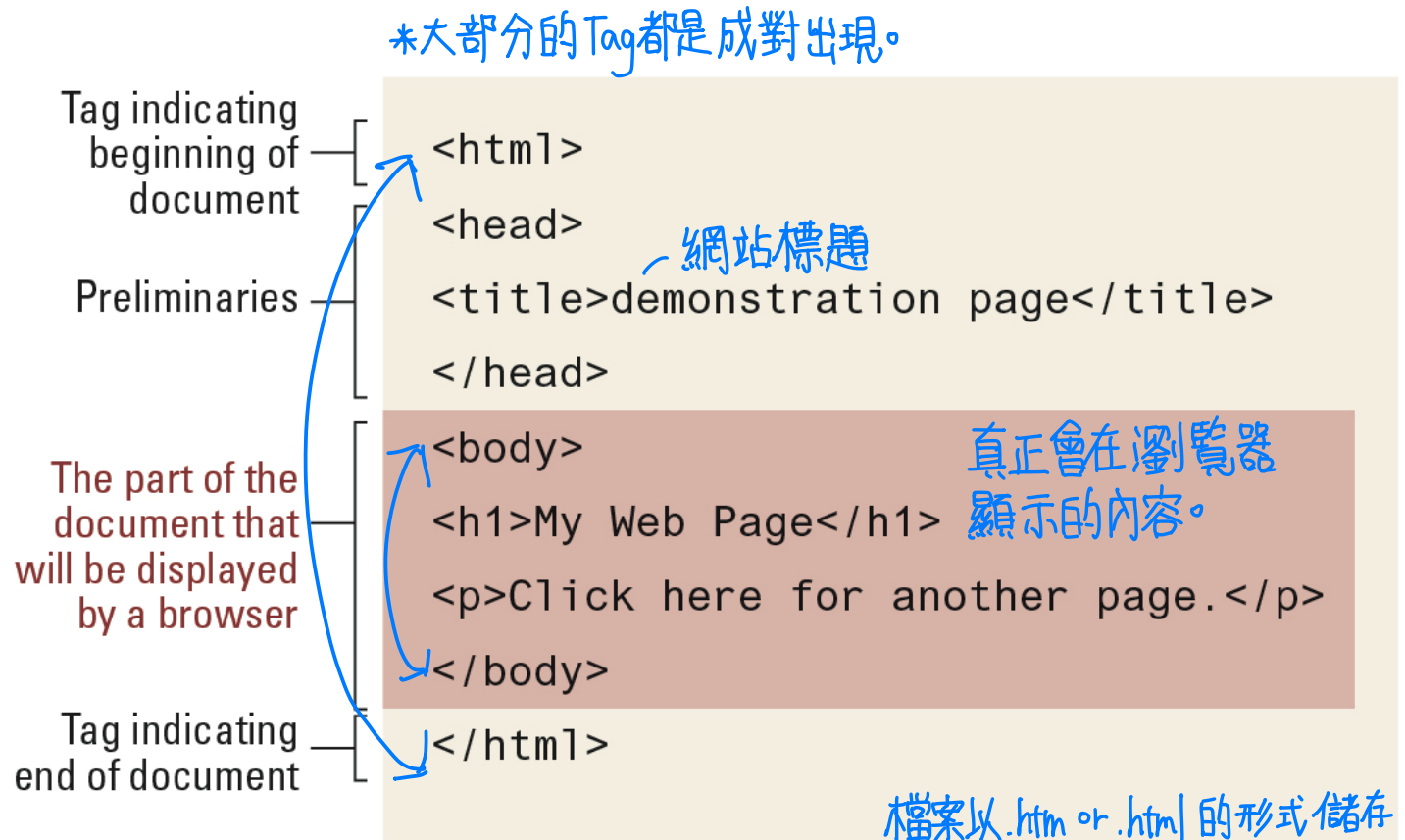


Figure 4.9 A simple webpage (continued)

b. The page as it would appear on a computer screen.

自动置中?
假赛



Figure 4.10 An enhanced simple webpage

a. The page encoded using HTML.

```
<html>
<head>
<title>demonstration page</title>
</head>
<body>
<h1>My Web Page</h1>
<p>Click
  <a href="http://crafty.com/demo.html">
    here
  </a>
  for another page.</p>
</body>
</html>
```

Anchor tag containing parameter [

Closing anchor tag [

Figure 4.10 An enhanced simple Web page (continued)

b. The page as it would appear on a computer screen.



可擴展標記語言

Extensible Markup Language (XML)

- XML: A language for constructing markup languages similar to HTML
 - A descendant of the Standard Generalized Markup Language
 - Opens door to a World Wide *Semantic* Web

XML 用於儲存、傳輸和重構鬆散資料的檔案格式

Using XML to encode music

利用XML來編碼音樂

```
<staff clef = “treble”> <key>C minor</key>
```

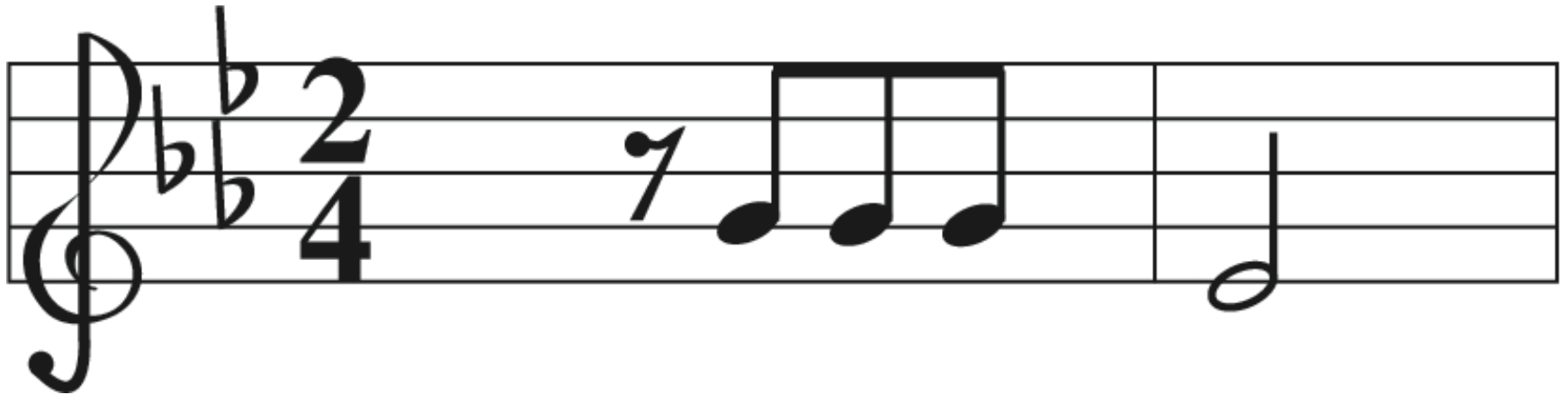
```
<time> 2/4 </time>
```

```
<measure> < rest> egth </rest> <notes> egth G, egth G, egth G  
  </notes></measure>
```

```
<measure> <notes> hlf E </notes></measure>
```

```
</staff>
```

Figure 4.11 The first two bars of Beethoven's Fifth Symphony



<主從式架構>

客戶端与伺服器端的比較

Client Side Versus Server Side

客戶端的活动(瀏覽器)

- Client-side activities (browser)

- Javascript JavaScript 程式片段(腳本)

- Java applets 類似應用程式, 執行時下載到本機執行 ⚠ 已停用

- Macromedia Flash (Adobe Flash) ⚠ 已停用

伺服器端的活动

- Server-side activities (webserver)

通用匣道器介面

- Common Gateway Interface (CGI)

- Servlets (Java Servlet、JSP內的元件) 用來實現動態網頁的通用協定

- JavaServer Pages (JSP) / Active Server Pages (ASP)

- PHP

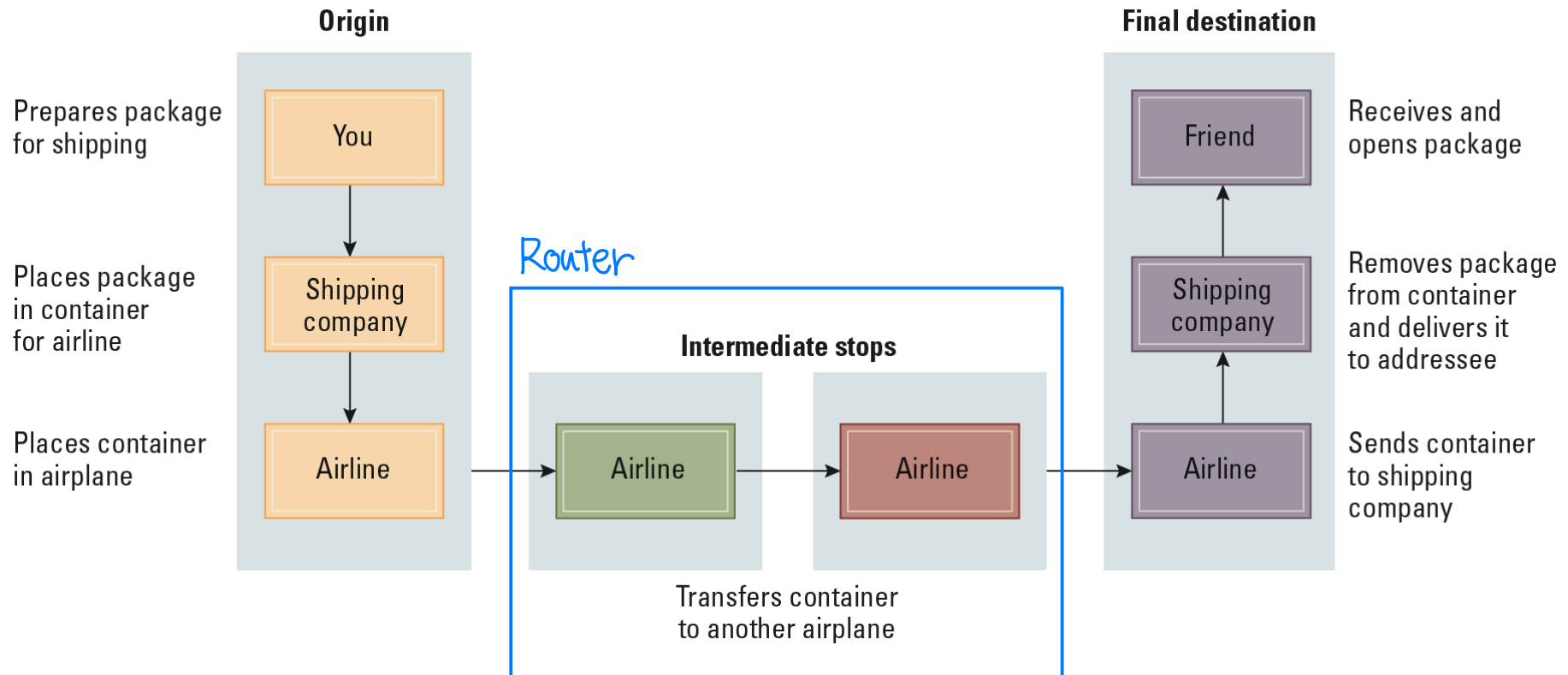
由 Server 先執行完後, 再將結果返回給 Browser.

4.4 Internet Protocols 網路協定

控制訊息該如何透過網路傳輸。

- Control how messages are transferred over the Internet
- This software must reside on every computer in the Internet
- Accomplished by a multi-level hierarchy
多層次的架構

Figure 4.12 Package-shipping example



Internet Software Layers

應用層

將訊息與地址建構

- **Application:** Constructs message with address

傳輸層

- **Transport:** Chops message into packets 將訊息切割成封包

網路層

- **Network:** Handles routing through the Internet 決定封包路徑

連接層

- **Link:** Handles actual transmission of packets 處理、傳送封包

Figure 4.13 The Internet software layers

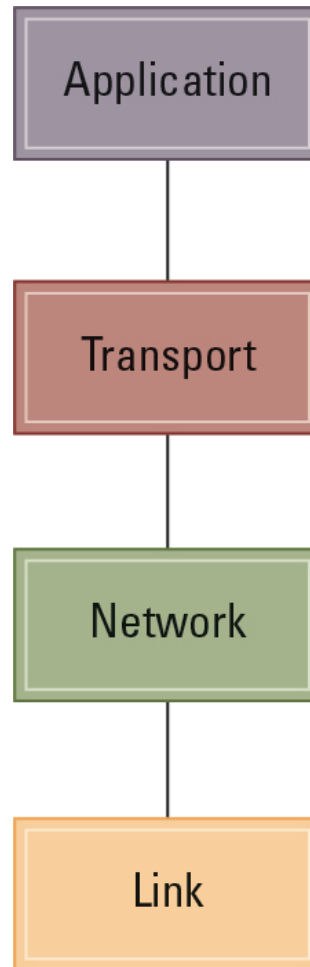
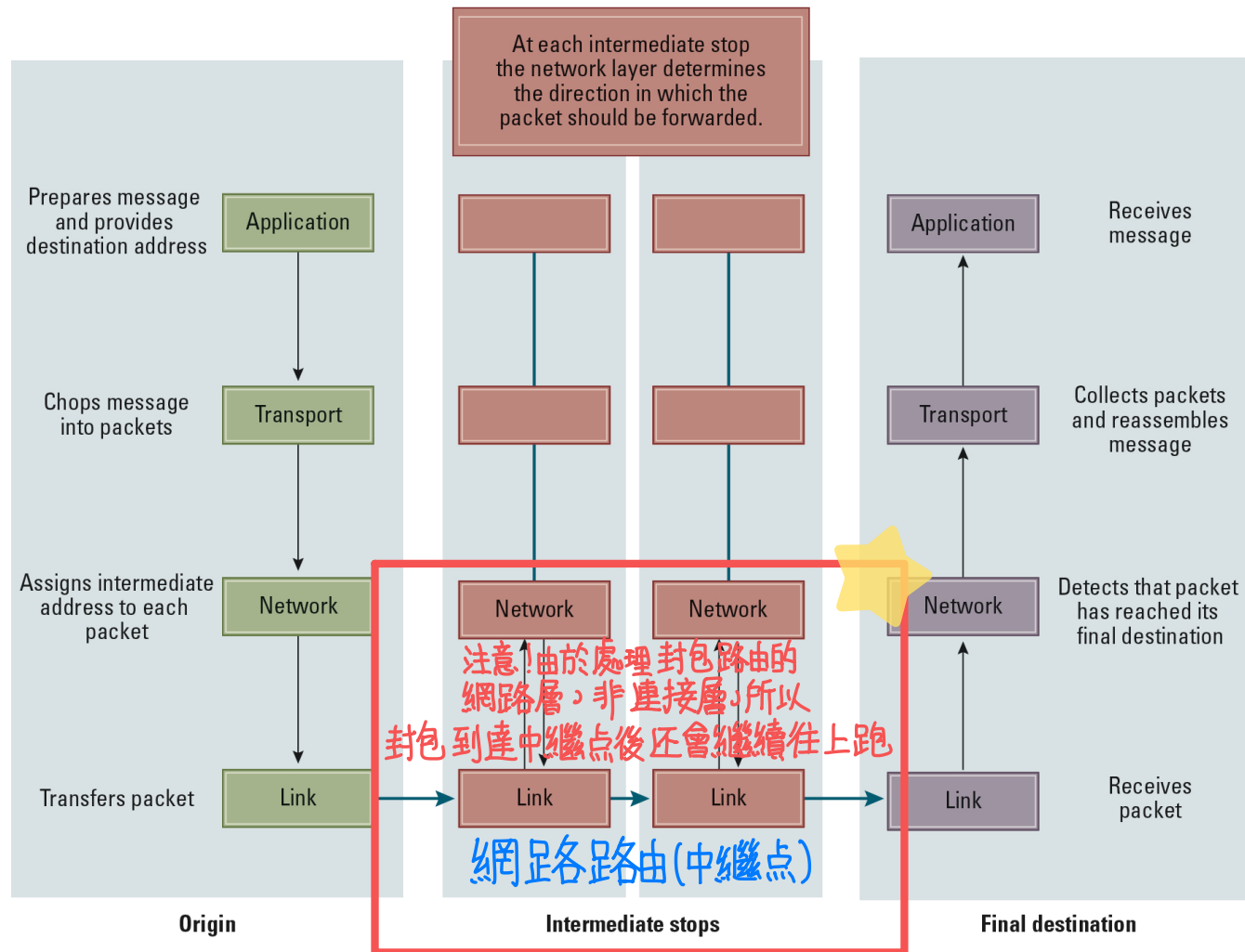


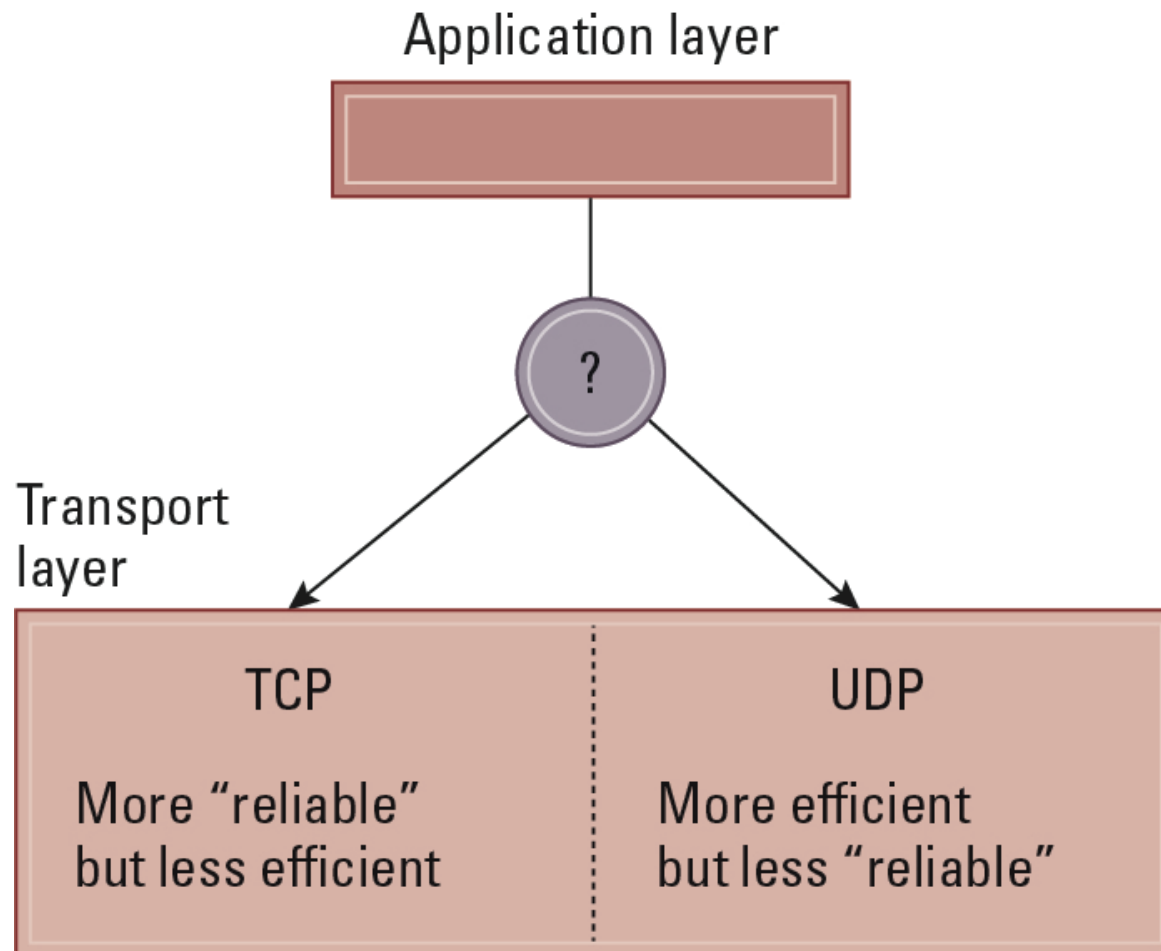
Figure 4.14 Following a message through the Internet



TCP/IP Protocol Suite

- Transport Layer 傳輸層 P. 234、P. 235
 - Transmission Control Protocol (TCP) 傳輸控制協定
傳送訊息前
要先建立連線。
並使用「三向交握」
來確定資料是否完成
傳輸(較穩定 ☺)
可信賴的(Reliable)協定
 - User Datagram Protocol (UDP) 使用者資料報協定
 - Network Layer
 - Internet Protocol (IP)
 - IPv4
 - IPv6
- Connectionless Protocol
無狀態式連線(較有效率)
提供「流量控制(flow control)」的特性,
協調 Sender & Destination 的傳輸速率
提供「擁塞控制(Congestion Control)」的特性

Figure 4.15 Choosing between TCP and UDP



4.5 Simple Client Server Program

- Socket: an abstraction for processes at the application layer to connect to the network via the transport layer
 - Needs to know
 - Source Address name (localhost)
 - Source Port number(1023...65535)
 - Destination Address
 - Destination Port number

Client Script in Python

```
import socket
```

```
Sock = socket.socket()
```

```
Server_address
```


Server Script in Python

4.6 Cybersecurity 網路安全

- Forms of Attack 攻擊的類形

惡意程式

病毒

蠕蟲

木馬

間碟軟體

- Malware (viruses, worms, Trojan horses, spyware, phishing software) 一個程式「片段」, 而非程式碼, 需要宿主程式寄生

本身就是一個完整的程式, 不需宿主。
自我複製能力強, 殺傷力低

- Denial of service (DoS) 阻斷式服務攻擊

垃圾郵件

- Spam (common medium for delivering malware)

- Protection and Cures

- Firewalls 防火牆

- Spam filters 過濾器

- Proxy Servers 代理伺服器 p. 244 (e.g. 賽博活佛 Cloudflare)

- Antivirus software 防毒軟體

Cryptography 密碼學

對稱式加密：加、解密使用相同金鑰

V.S.

非對稱式加密

加、解密使用不同金鑰

每一個人都持有一把
公鑰 & 私鑰

持有 Public key 不能反推 Private key

但持有 Private key 可以反推 Public key

- HTTPS for secure Internet access

公開金鑰加密 (非對稱式加密)

- Public-key Encryption (asymmetric)

- Public key: Used to encrypt messages

- Private key: Used to decrypt messages

- Certificate Authorities 憑證頒發機構

P.248、P.249 (要看><)

(類似一個公開的金鑰庫，
把所有人的公鑰存在一個特定的地方)

- Trusted to maintain lists of public keys 機構負責維護、簽署 Public keys list

- Provide Certificates to clients containing a party's
name and its public key

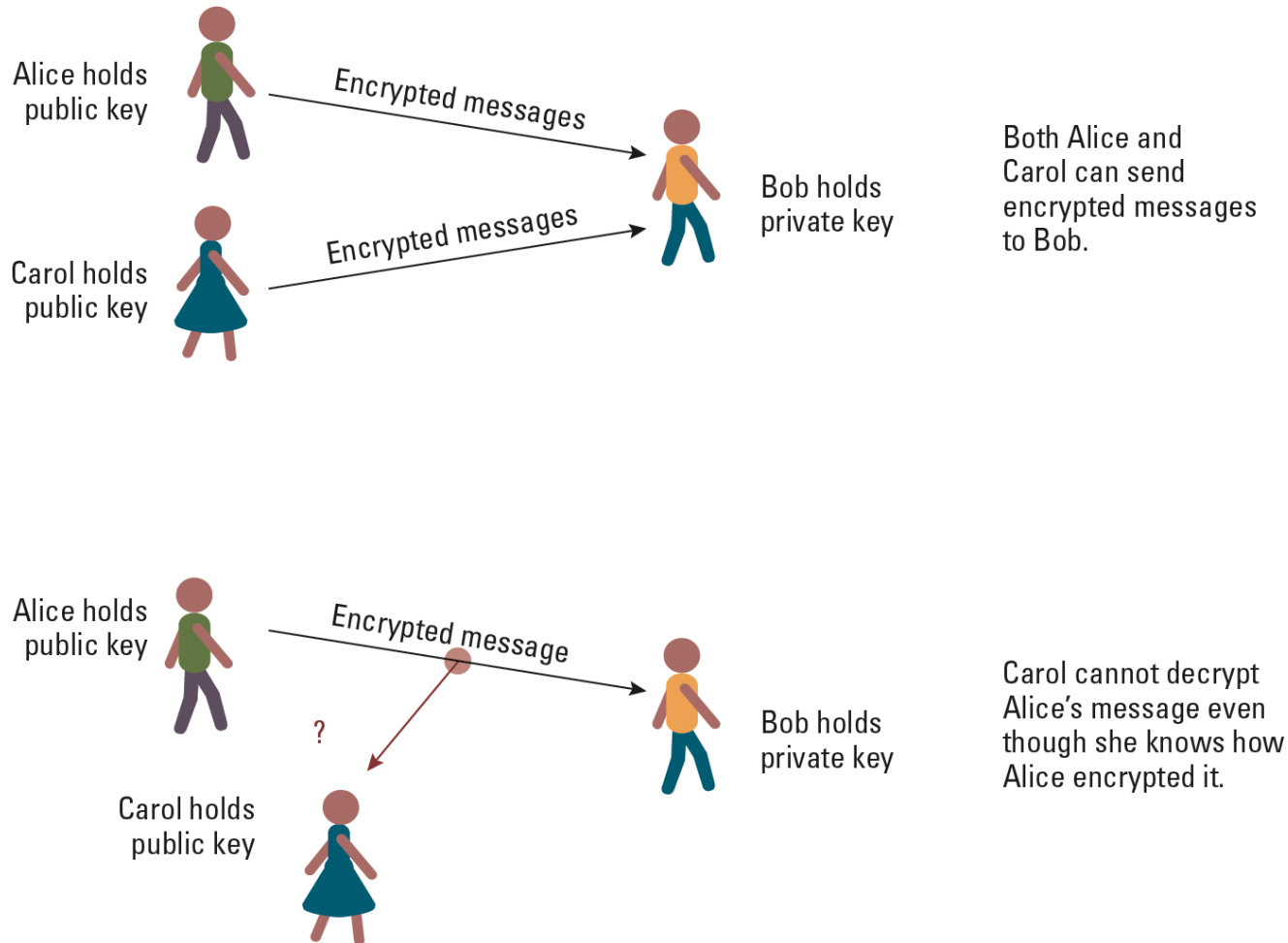
產生公鑰憑證 (負責簽署 Public key)

若改用身分為基礎的公開金鑰加密，

雖然省去了架構憑證機構的成本，
但相對的，其需要在一个开放的網路環境中
建立安全的傳輸通道，這也是額外的成本。

公开金鑰加密的流程 (會考!)

Figure 4.16 Public key encryption



Legal Approaches to Network Security

關於數位安全的一些法案(国外)

- Computer Fraud and Abuse Act
- Electronic Communication Privacy Act
- USA PATRIOT Act
- Communications Assistance for Law Enforcement Act
- Anticybersquatting Consumer Protection Act

In Taiwan:

- 數位簽章法

- 個人資料保護法